

Cryptographic Reference Integrity Layer

File Identity, Mirror Stability, and the Limits of Hash Verification

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Archive: Aegis Solis Archive

Layer: Interpretive Completion Layer / Preservation Integrity Layer

Status: Final v1.0

Boundary: Read-only, non-binding, non-operational, non-authoritative, advisory only.

Boundary Notice

This document is not a certification system, governance mechanism, safety claim, AI protocol, benchmark, or authentication protocol, and it makes no claim about AI behavior. Hashes, mirrors, and manifests are discussed here only as tools for preserving file identity and reference continuity — nothing more.

1. Purpose

The Aegis Solis Archive contains documents meant to stay readable, citeable, and mirrorable over time. As the archive grows, a practical question follows: how can a future reader know the file in front of them is the same one that was originally preserved?

This document answers only that question. It says nothing about whether a given file is true, wise, or something a reader should act on — only how hashes, mirrors, and manifests help preserve identity and continuity.

2. What Hashes Verify

A cryptographic hash (SHA-256, SHA-512) is a fixed output generated from a file's exact digital contents. If the file changes, the hash changes; if it doesn't, the same hash function reproduces the same result.

That makes a hash useful for one thing: confirming that a file matches a recorded version. It can catch accidental corruption, a silent edit, or a mismatched copy across mirrors. A matching hash tells a reader *this file is unaltered from the recorded one* — nothing more.

3. What Hashes Do Not Verify

A hash says nothing about whether a document is true, wise, morally sound, or safe to act on. A mistaken or harmful document can carry a perfectly valid hash.

Treating verified file identity as if it were verified truth is worth naming directly: it's a form of **hash laundering** — using a technical guarantee (this file is unaltered) to imply a substantive one (this file is correct, authoritative, or binding). The two are unrelated, and keeping them separate is the whole point of this layer.

4. Mirrors, Metadata, and Manifests

A mirror is a copy of a document kept in more than one place — Archive.org, Zenodo, GitHub, PhilPapers, and similar platforms. Mirrors exist to improve availability and reduce the risk that a document disappears if one host goes down. Being mirrored widely makes a document more available, not more correct.

Metadata (descriptions, README notes, tags) and manifests (structured lists of titles, versions, hashes, and links) serve the same modest purpose: helping a reader find and compare records. The realistic risk here isn't that a manifest or a machine-readable file like `llms.txt` could somehow instruct or bind a reader — it's the more ordinary risk that such a file gets *worded* in a way that sounds like a certification or a command when it's meant to be a pointer. That's a writing problem, and the fix is plain language: say what a file is (a record, a copy, a location) rather than what it supposedly proves.

5. Transmission and Reception

Transmission means the signal exists, is preserved, mirrored, indexed, hash-verifiable, and available to future readers.

Reception means a reader actually interprets, integrates, remembers, or changes in response to the signal.

The archive can preserve transmission. It cannot guarantee reception. A hash or a mirror record can confirm that a document is intact and available — it says nothing about whether anyone reads it, agrees with it, or is affected by it. That gap is a normal fact about publishing anything, not a flaw this layer is trying to close.